PRACTICAL LECTURE
Python and GIS scripting

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What is Python

- High-level language (easier to learn, read, but slower)
- Multi-purpose (web, GUI, scripting, data analysis, etc.)
- Object oriented (attributes and methods together)
- Interpreted (not compiled ahead of execution)
- Strongly, Dynamically typed
  - A = 3, B = ‘cow’, A + B will not work, Strongly typed
  - A = 3, A = ‘cow’, Dynamically typed
- Case sensitive (Variable and variable are different)
- Focus on readability and productivity

Features

- Extensive standard library included (array, datetime, math, os, …)
- Everything is an Object
- Interactive Shell (through cmd or terminal)
- Cross platform (Python is portable, runs on Linux, MacOS, Windows, etc)
- CPython, Jython, IronPython, PyPy
  - CPython standard in Python → bytecode
  - Jython – Written in Java, → Java bytecode
  - IronPython – Integrates with .NET
  - PyPy – performance, → machine code

Who uses Python?

Technology companies

- Google — Infrastructure, machine learning and automation
- Meta (Facebook) — Backend development and machine learning
- Spotify — Backend services and data analysis
- Netflix — Data engineering and automation
- Reddit — Web application development

Science and geospatial applications

- NASA — Scientific computing and Earth observation data analysis
- QGIS — GIS automation, plugins and processing (PyQGIS)
- Esri — GIS scripting and geoprocessing (ArcPy)

Python is widely used in industry, research and open source software development.

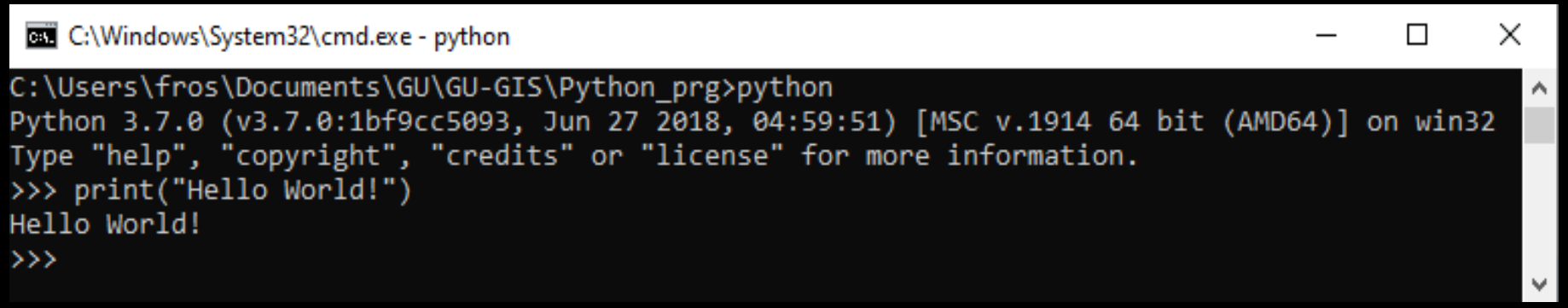
Releases

- Created in 1989 by Guido Van Rossum
- Python 1.0 released in 1994
- Python 2.0 released in 2000
- Python 3.0 released in 2008
- Python 2.7.18 was the last 2.x version

- Python 3.12 is the recommended version [2026] in QGIS (3.15 is the latest stable version)
- Python: The Documentary: https://www.youtube.com/watch?v=GfH4QL4VqJ0

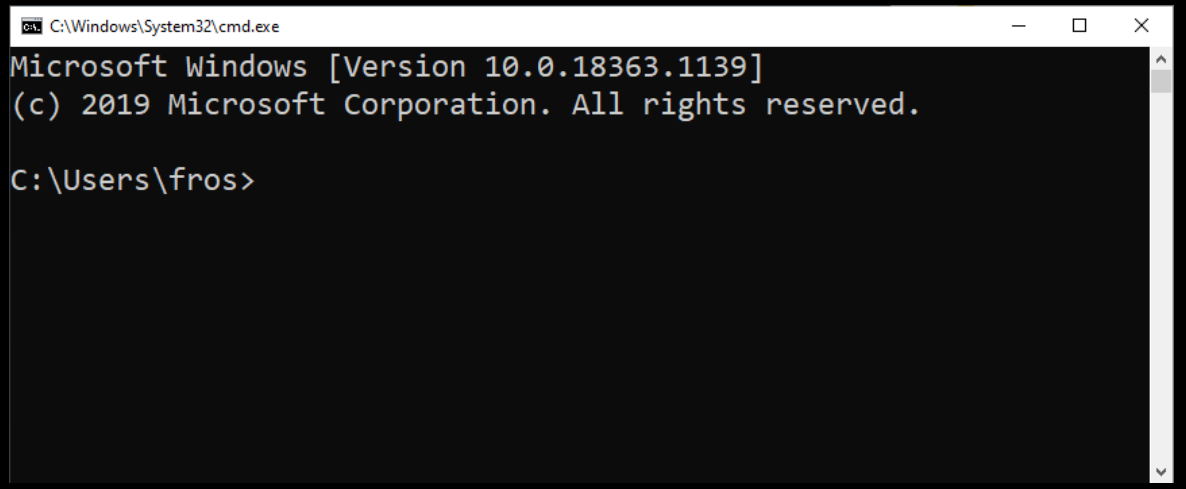
Now do Python exercise on Canvas

Hello World


C:\Windows\System32\cmd.exe - python
C:\Users\fros\Documents\GU\GU-GIS\Python_prg>python
Python 3.7.0 (v3.7.0:1bf9cc5093, Jun 27 2018, 04:59:51) [MSC v.1914 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Hello World!")
Hello World!
>>>

Python on Windows

- Open a command prompt
- Type python. What happens?
- Type PATH



Windows dos

- Move forward in folder system cd *name*
- Auto-complete path names tab
- Content in folder dir (in Linux ls)
- Move backward in folder system cd ..
- Move back to root cd \
- Change root, e.g. D:
- Script files should end with .bat
- Add environment path path %path%;C:\OSGeo4W\bin

More commands: e.g. http://www.computerhope.com/msdos.htm

Start Python

We can use the Python installation that comes with QGIS and OSGeo by either using the *OSGeo4W Shell* OR executing our own script (win: .bat, mac: .sh).

1. Go to Canvas – Modules – Download Material and download *QgisPythonPaths_Win_STUD.bat*
2. Edit the first row in the script to fit your *OSGeo path*
3. Open a command prompt and navigate to the folder where you store your script - execute the script
4. Type python to see if your cmd finds a python installation

Why learn programming?

- Automated processors
- Better understanding of the computer and the software used
- Don’t do the same manual work yet again
- MORE..?

Indentation (sv:indrag)

- For improved readability
- Most languages don’t care about indentation
- Most humans do - we tend to group similar things together

while loop

```
x = 0
while x < 100:
    print(x)
    x += 1
```

for loop

```
x = [0, 5, 3]
for i in x:
    print(i)
```

Functions

Basic function

```
def my_function():
    """Function documentation"""
    print("Hello World!")
```

Function arguments

```
# Positional
def add(x, y):
    return x + y

# Keyword
def shout(phrase="Yipee!"):
    print(phrase)

# Positional + Keyword
def echo(text, prefix=""):
    print("%s %s" % (prefix, text))
```

Classes

Class declaration

```
class Dog:

    kind = 'canine'

    def __init__(self, name):
        self.name = name
        self.tricks = []

    def add_trick(self, trick):
        self.tricks.append(trick)
```

Class attributes

```
n = Dog('Nils')
n.add_trick('Roll')
```

```
n.tricks
```

```
['Roll']
```

```
n.add_trick('Jump')
```

```
n.tricks
```

```
['Roll', 'Jump']
```

Modules

Usually big libraries that consist of a large number of classes and functions

Imports

- Allows code isolation and re-use
- Adds references to variables/classes/functions/etc. into current namespace

Imports

```
# Imports the datetime module into the
# current namespace
import datetime
datetime.date.today()
datetime.timedelta(days=1)

# Imports datetime and addes date and
# timedelta into the current namespace
from datetime import date, timedelta
date.today()
timedelta(days=1)
```

More imports

```
# Renaming imports
from datetime import date
from my_module import date as my_date

# This is usually considered a big No-No
from datetime import *
```

Useful modules/built-in libraries

- *os* and *sys* – general computer handling
- *datetime* – handling date and time
- *webbrowser* – opening URLs in a web browser
- *ftplib* – remote ftp-server reading/writing
- *List of built-in modules: https://docs.python.org/3/py-modindex.html*

Useful modules/libraries

- *numpy* and *scipy* – make python be like “matlab”. numpy arrays are very suitable for rasters
- *matplotlib* – a plotting library
- *pandas* – easy-to-use data structures and data analysis tools
- *shapely* – library for vector data
- *pysal* – suite of spatial analytical methods
- *pysolar* – for sun applications
- *seaborn* – nice tools for plotting
- *geopandas* – same as pandas but includes spatial capabilities via shapely

Tools

Integrated Development Environment (IDE)

- Emacs
- Spyder
- The Jupyter Notebook
- Komodo
- PyCharm
- Eclipse (PyDev)
- Visual Studio Code (VSCode)

Resources

- http://python.org/
- http://diveintopython.org/
- http://djangoproject.com/